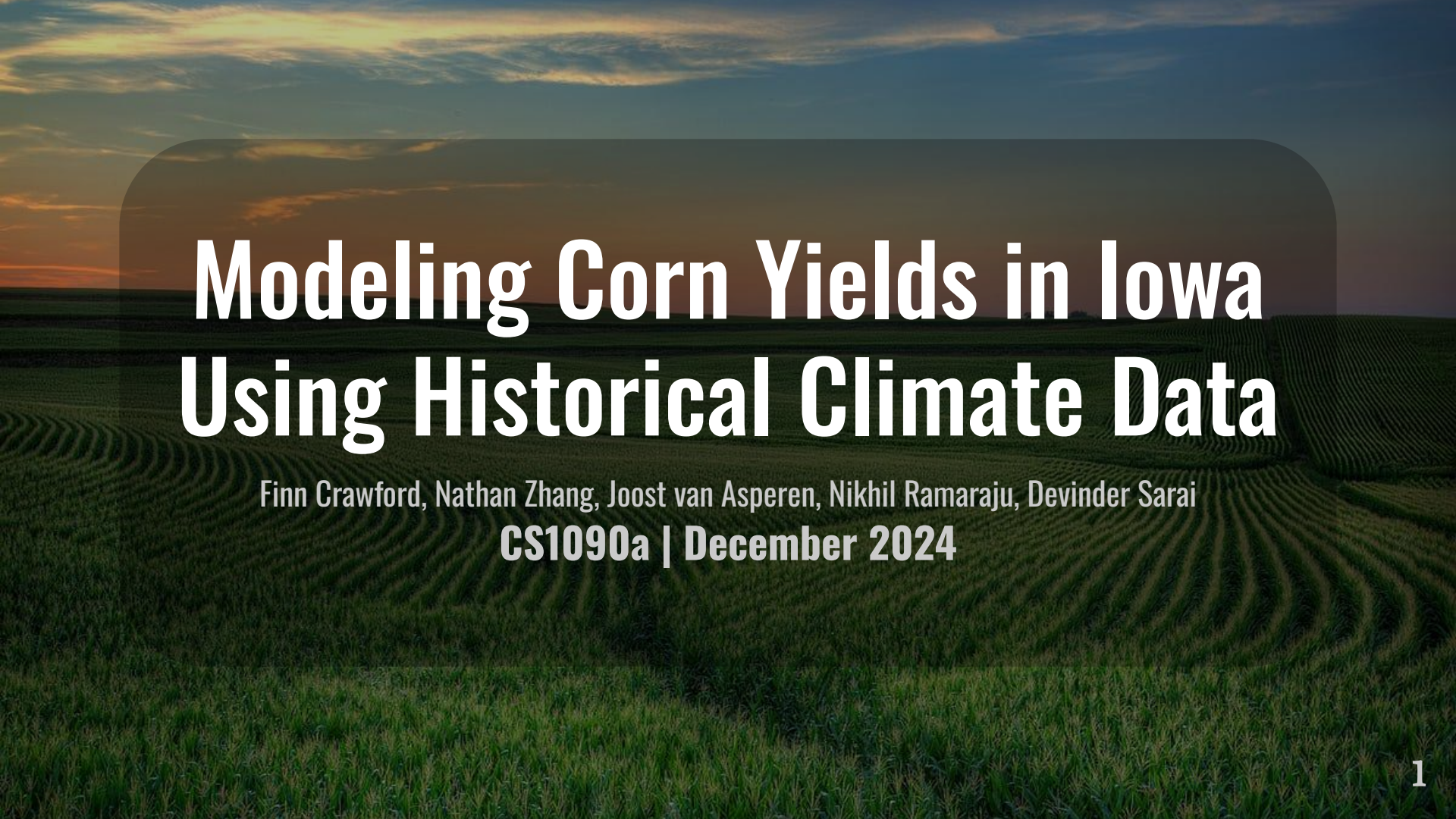




Modeling Corn Yields in Iowa Using Historical Climate Data

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Research Question:

**Can climate data be used to
predict corn yields in Iowa?**



Motivation

- **Climate** affecting regions in unpredictable ways.
- Disrupting the conditions needed for efficient large-scale crop production.
- Models to understand and quantify the relationship between climate and **crop yield**.
- Growing human population requires increased agricultural yields.
- Can agricultural output keep up with growing demand despite climate change?

Data Sources

Climate Data (NOAA)



- Daily climate measurements 1926–2022
- Includes maximum/minimum temperature (°C) and precipitation (in)
- Ten stations across Iowa
- n = 356598

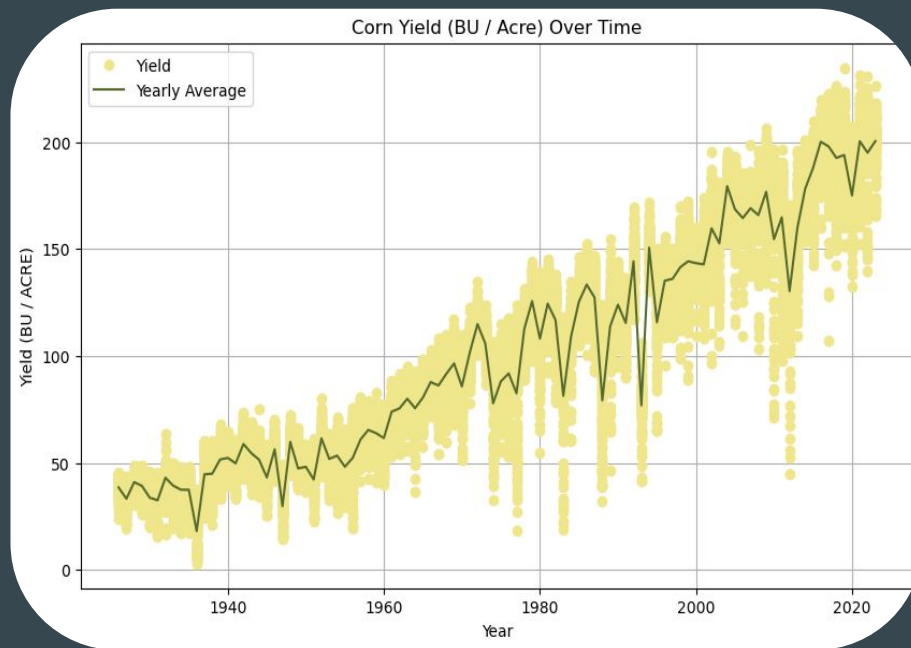
Agricultural Data (USDA)



- Annual county-level corn yields 1926–2022
- Measured in Bushels/Acre
- 98 counties across Iowa
- n = 9658

Exploratory Data Analysis

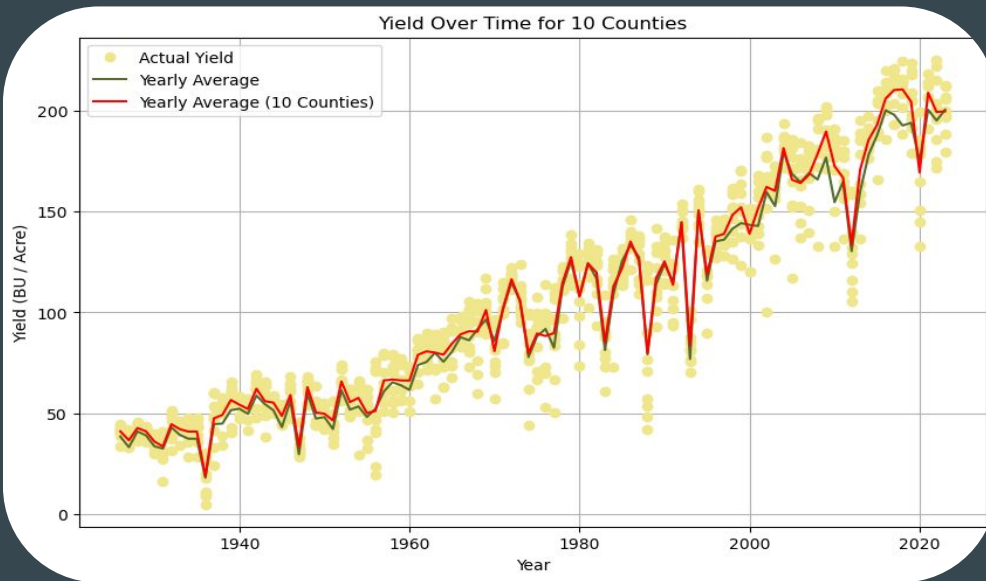
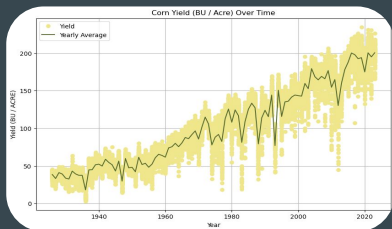
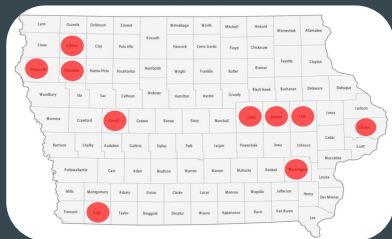
- Clear trend of improving yield over time (~30 → 170 Bu/Acre)
- Increase likely due to improvements in agricultural technology (Dimitri et al., 2005)
- Using 'year' as an additional feature to proxy for improvements in technology



Data Preprocessing



Weather Station Coverage



- Cut down crop yield data to just 10 the counties containing weather stations
- Observed minimal change in yearly average trend between original and cropped data
- Encode the 10 selected countries into binary columns using **OneHotEncoding**

Climate Parameters

1. Constrain data to Growing Season (May 1 – October 15th)

PRCP

TMAX

TMIN

2. Create custom parameters using existing data:

GDD

“Growing Degree Days”

$$(T_{\text{Max}} + T_{\text{min}})/2 - 50^{\circ}\text{F}$$

A measure of heat accumulation used to estimate plant growth

Heat Shock

of days with:
 $T_{\text{max}} > 90^{\circ}\text{F}$

Cold Shock

of days with:
 $T_{\text{min}} < 32^{\circ}\text{F}$

PRCP

$\Sigma(\text{PRCP})$ during growing season

3. Create interaction term **GDD x PRCP**

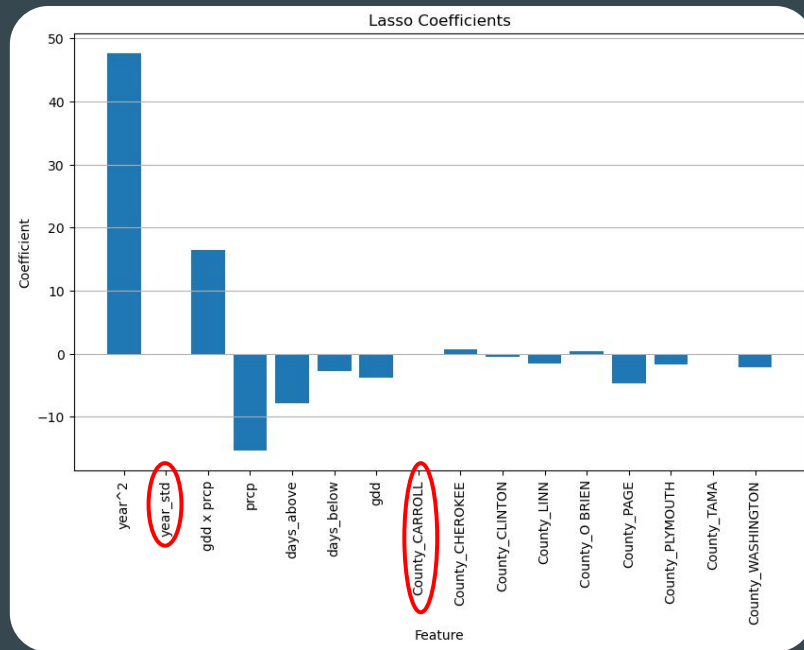
4. Standardize final terms

Baseline Model (LASSO Regression)



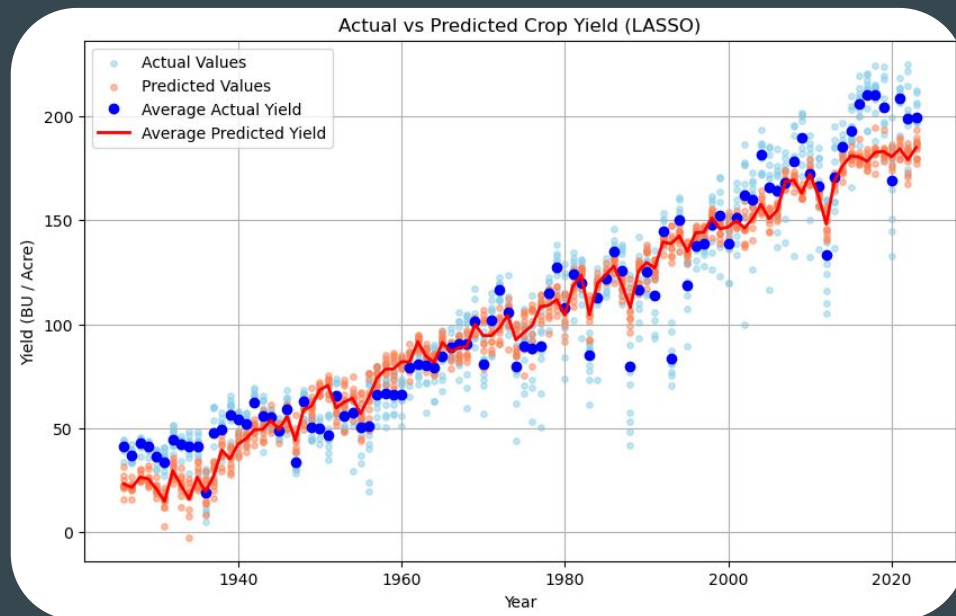
LASSO Coefficients

- Most important predictor is **Year²**
 - Improvements in technology have led to the greatest increase in yield
- **Year** parameter shrunk to 0
 - High degree of multicollinearity with **Year²**
- Environmental predictors such as **gdd x prcp** and **prcp** have high coefficients as well
- **County_CARROLL** also shrunk to 0
 - Likely a High degree of similarity with the reference county (**BENTON**)

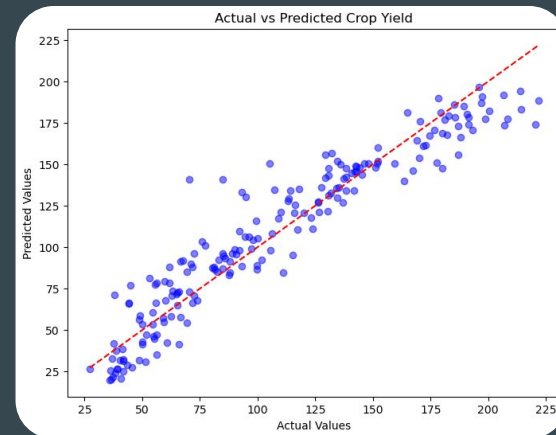


(cv=5, $\alpha=0.05$)

LASSO Results: Prediction

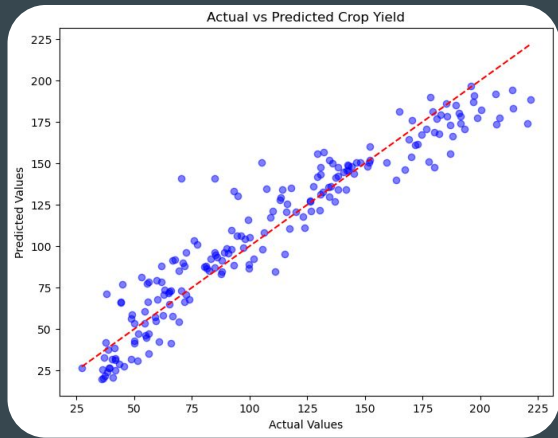
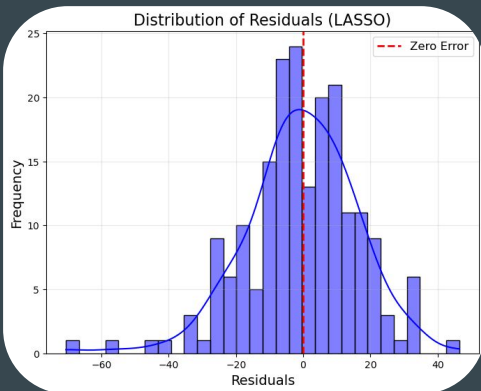
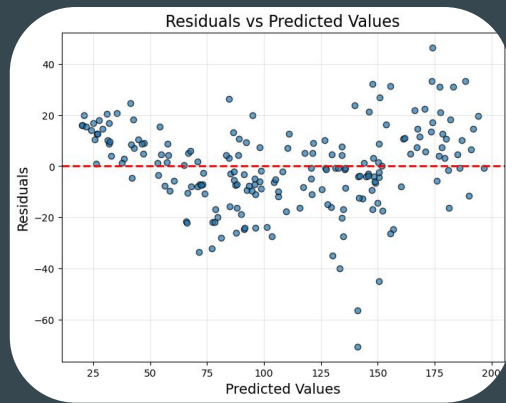


- Model achieves an R^2 value of ~ 0.9 , meaning the model accounts for roughly 90% of observed variance
- However, the predictions appear more stable than reality, and the model seems unable to capture some of the 'dips' in yield



MSE: 272.51
R-squared: 0.898

LASSO Results: Error



- Error appears somewhat **heteroskedastic**, and residual distribution is **left-skewed**, suggesting the model is systematically failing to predict 'dips' in yield at higher values
- In this case, a **Random Forest** model may be a good next step to capture potential non-linear relationships driving this discrepancy

MSE: 272.51
R-squared: 0.898

Final Model

(Random Forest Regression)



Model Pipeline

Hyperparameter Tuning

- Define a **parameter grid** using specified values
- Find optimal hyperparameter combination using **GridSearchCV** five-fold cross-validation

Model Training

- Train a **single decision tree** using optimal parameters
- Train a **random forest regression** model with the same parameters

Model Evaluation

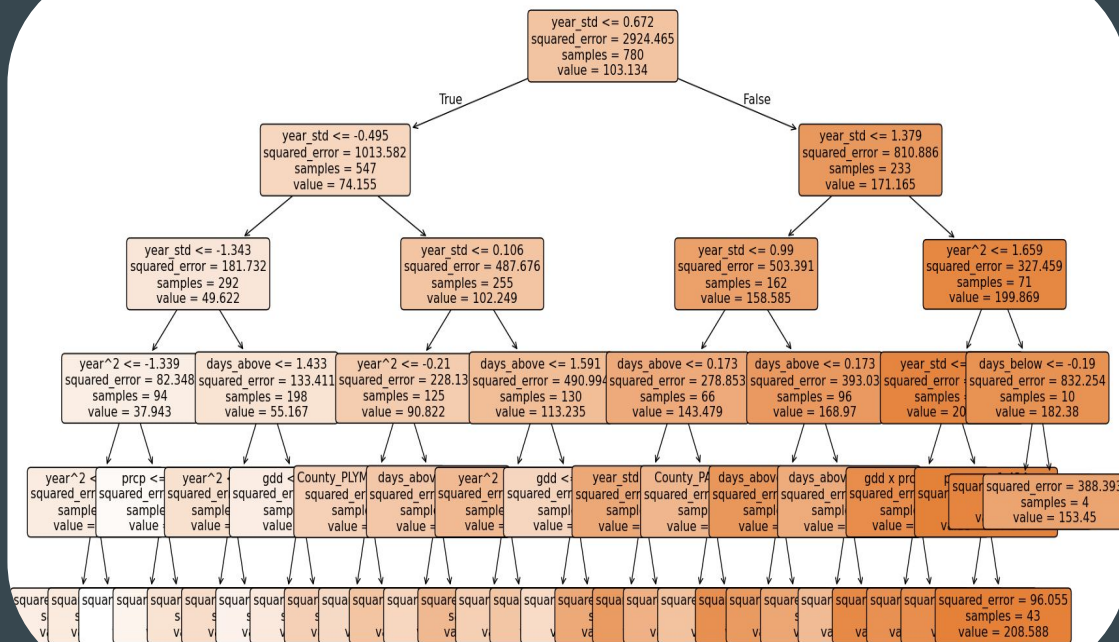
- Calculate **MSE** and **R²** for both models to assess whether the increased complexity of RF is justified
- Use single DT to interpret parameter relationships

Visualization/Analysis

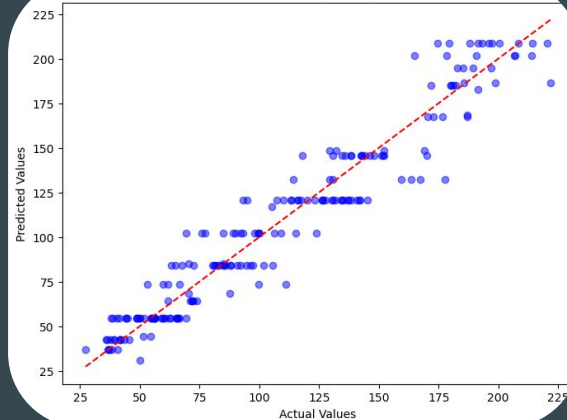
- Visualize RF feature importances
- Plot RF predictions against annual yield trend

Single Decision Tree

Regression Decision Tree

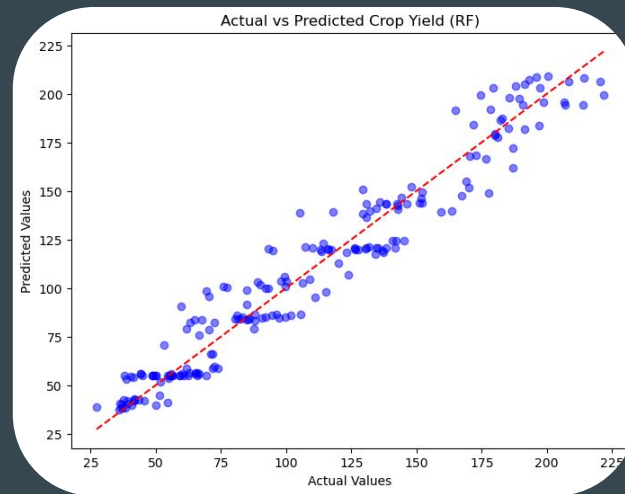
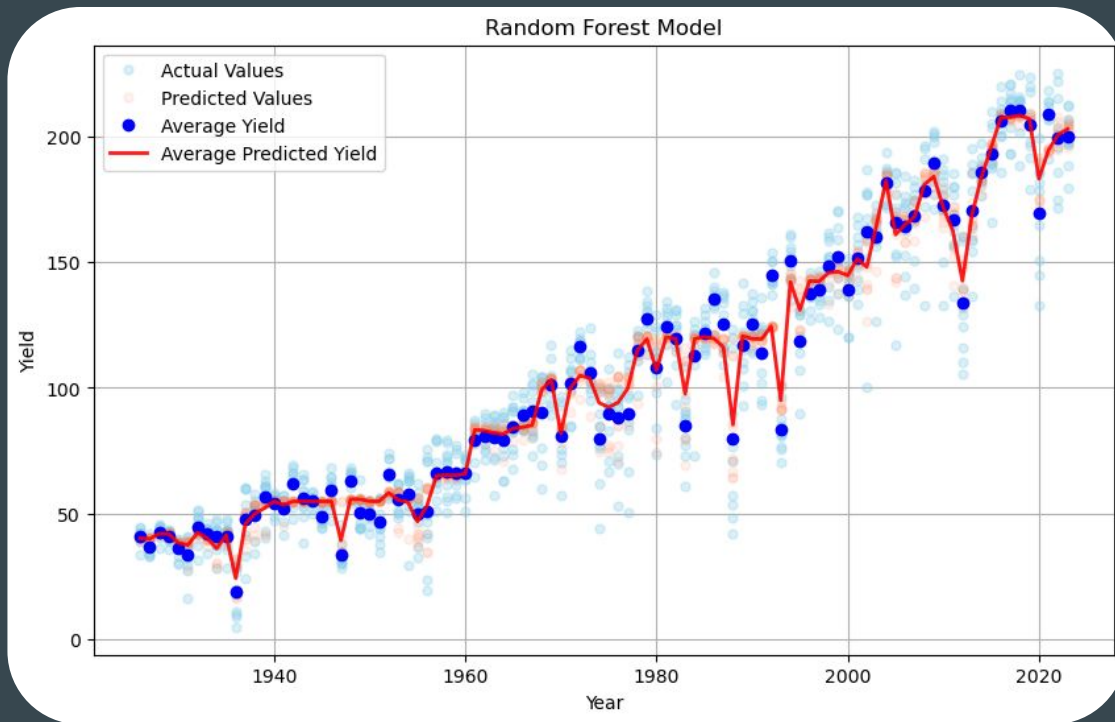


Actual vs Predicted Crop Yield (Decision Tree)



MSE: 190.46
R-squared: 0.929

Random Forest



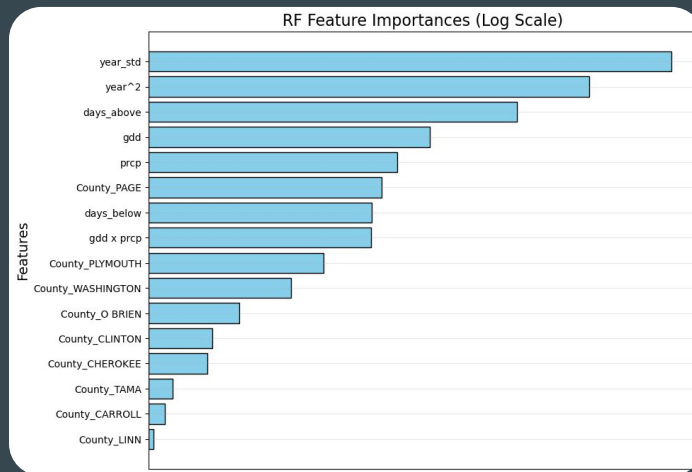
DT
MSE: 190.46
 R^2 : 0.929

RF
MSE: 123.54
 R^2 : 0.954

Feature Importances (RF)

- **Year** and **Year²** are once again the most important features, indicating a strong reliance on technological advancement
- **Days_above** is the next most important feature, demonstrating the potential negative implications of heat shock on corn crops
- **GDD** and **prcp** then follow, showing that precipitation and temperature still play key roles in crop yield

```
year_std: 0.8216918855286927
year^2: 0.13777640983272774
days_above: 0.029065831790568313
gdd: 0.004382772567653218
prcp: 0.0021843109857726368
County_PAGE: 0.0015498456473096531
days_below: 0.0012616578560102189
gdd x prcp: 0.001238265924494535
County_PLYMOUTH: 0.000441317647250491
County_WASHINGTON: 0.00021847054219067489
County_O_BRIEN: 7.13976925926274e-05
County_CLINTON: 4.007092216829735e-05
County_CHEROKEE: 3.560568623646193e-05
County_TAMA: 1.6865623605664803e-05
County_CARROLL: 1.4226613682776709e-05
County_LINN: 1.1065139044097589e-05
```

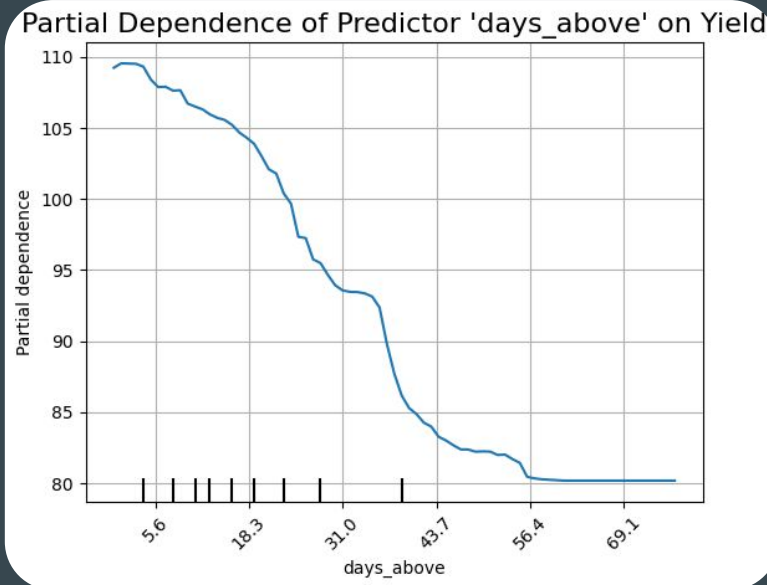


Sensitivity Analysis



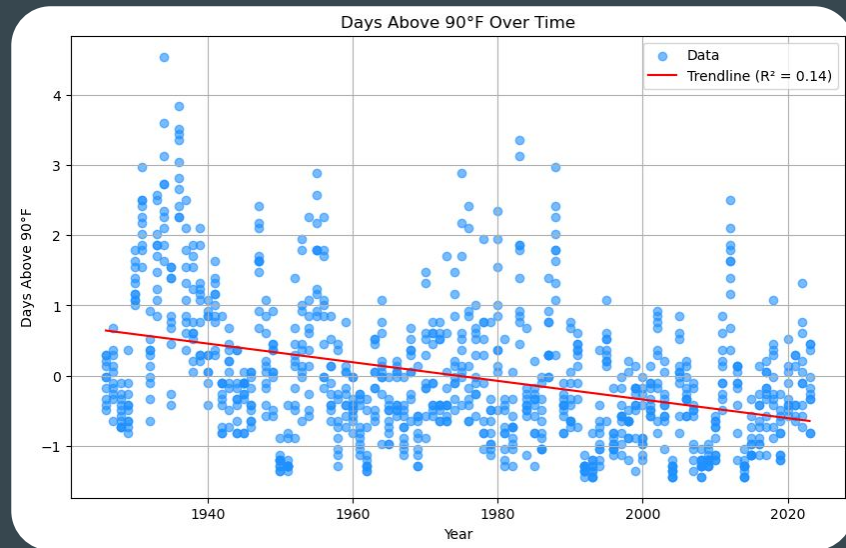
Partial Dependence Analysis

- Given **days_above** was the most important climate parameter in the RF mode, we elected to perform partial dependence analysis to further quantify its importance
- We find that the predicted yield changes greatly as we change the **days_above** variable
- This may be alarming, as climate change will likely increase this predictor



A Word to Stakeholders

- Although our results from the partial dependence analysis may seem frightening, we see from the plot on the right that **days_above** exhibits no strong trends over the past century ($R^2 = 0.14$).
- Our trendline shows a weak decrease in the **days_above** predictor over time, in line with a slight decrease in **GDD** over the same period.
- However, using information from our previous analysis, stakeholders should still be wary about the implications of temperature increase on crop yield.



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Thank You

